

Toward Thirty Independent Rational Points on an Elliptic Curve over $\mathbf{Q}$

A certificate-first search, exact obstructions, and cyclic cubic trace codes

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Abstract

We study the constructive problem of exhibiting an elliptic curve over $\mathbf{Q}$ with at least thirty independent rational points. The objective is not an impressive analytic-rank score but a final, independently checkable lower- rank certificate: one nonsingular Weierstrass equation, thirty complete rational points, exact equation checks, a rigorous independence proof, a canonical-height matrix, saturation data, provenance, hashes, and a second computer-algebra verification.

The current public rank-29 record is reproduced as the baseline. We then separate several obstructions that are often conflated: generic lattice capacity, arithmetic descent, visibility of low-height points, auxiliary-base genus, specialization dependence, and certification cost. Exact results include a branch-genus obstruction for multiquadratic packets, a principal- divisor obstruction to one-function thirty-point constructions, capacity bounds for torsion-friendly rational elliptic surfaces, and rigidity of elliptic fibrations when $\chi(\mathcal{O}_S) \neq 2$.

The principal new setting is a cyclic cubic genus-one pullback of a split E_8 rational elliptic surface. A degree-three elliptic base is the first low-genus base change capable of geometric Mordell–Weil rank thirty. We classify genus-one trisection classes by shells of E_8 , prove that every cyclic trisection orbit projects to an $A_2(2)$ plane independently of its intersection with the zero section, and identify an exact ternary trace code in $E_8/3E_8$. The minimal norm-six shell contains 6720 classes, but an integral projected packet is confined to one of 2240 maximal totally isotropic four-spaces, each containing only 240 minimal classes. This reduces the decisive search to a finite incidence problem: find eleven Hermitian-independent cyclic trisection orbits defining the same positive- rank genus-one cover. No rank-30 curve is claimed in this draft.

Truth status. The baseline proves $\text{rank } E(\mathbf{Q}) \geq 29$ for the published record curve. The new results are intermediate theorems and exact search reductions. They do not prove that rank 30 exists, and they do not prove a universal upper bound of 29.

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1 Exact formulation and reproduced baseline

1.1 The theorem that would solve the challenge

The required theorem is only a lower bound.

Theorem 1.1 (Target certificate). *There exists a nonsingular elliptic curve $E/\mathbf{Q}$ and rational points $P_1, \dots, P_{30} \in E(\mathbf{Q})$ such that*

$$\sum_{i=1}^{30} n_i P_i = O, \quad n_i \in \mathbf{Z},$$

implies $n_1 = \dots = n_{30} = 0$.

Exact rank 30 is optional. An upper bound, analytic or arithmetic, is not needed once thirty listed points have been proved independent.

1.2 Rank terminology

The following quantities must not be interchanged.

Quantity	Meaning
Algebraic rank	$\text{rank}_{\mathbf{Z}} E(\mathbf{Q})$.
Analytic rank	$\text{ord}_{s=1} L(E, s)$. Equality with algebraic rank is the Birch–Swinnerton-Dyer conjecture in general.
ℓ -Selmer rank	The dimension of the Selmer group after subtracting rational ℓ -torsion. It equals the Mordell–Weil rank plus a possible $\text{Sha}[\ell]$ contribution.
Generic rank	$\text{rank } E(\mathbf{Q}(t))$ for an elliptic surface over $\mathbf{P}^1$.
Specialization rank	$\text{rank } E_{t_0}(\mathbf{Q})$ for a nonsingular rational fibre.

Quantity	Meaning
Rank jump	The rank gained beyond the specialized generic subgroup.
Visible rank	The rank of the subgroup generated by points actually found.
Saturated rank	The rank of a subgroup together with a proof that no hidden prime division enlarges it inside its rational span.
Exact rank	A matching rigorous lower and upper bound.
Conditional upper bound	An upper bound depending explicitly on GRH, BSD, or another stated conjecture.

For every prime ℓ , the exact sequence

$$0 \longrightarrow E(\mathbf{Q})/\ell E(\mathbf{Q}) \longrightarrow \text{Sel}^{(\ell)}(E/\mathbf{Q}) \longrightarrow \text{Sha}(E/\mathbf{Q})[\ell] \longrightarrow 0$$

explains why a large Selmer rank is not a lower bound for the Mordell–Weil rank.

1.3 The current rank-29 record

The reproduced baseline uses the integral generalized Weierstrass equation

$$E_{29} : \quad y^2 + xy = x^3 - Ax + B,$$

where

$$A = 27006183241630922218434652145297453784768054621836357954737385,$$

$$B = 552580585513423764757366995911181918215210670325350796083724047791494132777161734256367214.$$

The repository stores the complete twenty-nine rational points in `points.json`.

The dependency-free verifier performs exact substitution, checks nonsingularity, and proves trivial rational torsion using good reductions

$$\#E(\mathbf{F}_{19}) = 28, \quad \#E(\mathbf{F}_{23}) = 32, \quad \#E(\mathbf{F}_{67}) = 83.$$

Their greatest common divisor is one.

For independence, the points are mapped to

$$\prod_{p \in S} E(\mathbf{F}_p)/2E(\mathbf{F}_p)$$

for twenty-two good primes. The resulting binary matrix has rank twenty-nine. If an integral relation existed, dividing its coefficients by their largest common power of two would produce a nonzero binary relation, a contradiction. Thus

$$\boxed{\text{rank } E_{29}(\mathbf{Q}) \geq 29}$$

unconditionally.

Independent SageMath and Magma computations reproduce the canonical-height matrix and a nonzero regulator, and the listed subgroup is tested for p -saturation for every prime $p < 4096$. These computations strengthen the certificate but are not substitutes for the exact finite-reduction proof. The published exact-rank upper statement remains clearly marked as conditional.

1.4 Certificate-first promotion levels

Search objects are classified as follows:

Level	Requirement
0	arithmetic score only; no rank claim,
1	at least 25 verified independent points,
2	at least 28 verified independent points,
3	29 independent points plus a distinguished unresolved covering or point,
4	thirty putative points, not yet rigorously independent,
5	thirty rigorously independent rational points.

Only Level 5 is a record.

2 Obstruction atlas

The observed difficulty of finding one additional point is not a single phenomenon. It is the product of several mathematically different obstructions.

2.1 Surface-lattice capacity

For a relatively minimal elliptic surface $\pi : S \rightarrow \mathbf{P}^1$ with section, Shioda–Tate gives

$$\text{rank MW}(S) = \rho(S) - 2 - \sum_v (m_v - 1),$$

where m_v is the number of irreducible components of the fibre at v . For arithmetic genus $d = \chi(\mathcal{O}_S)$ one has

$$h^{1,1}(S) = 10d, \quad \text{rank MW}(S) \leq 10d - 2.$$

Thus a rational elliptic surface has generic rank at most 8, a K3 elliptic surface at most 18 geometrically, and an arithmetic-genus-three surface at most 28. These are generic-rank bounds; they do not bound a single specialized fibre over $\mathbf{Q}$.

Every reducible fibre consumes Picard rank through its root lattice. This is the *reducible-fibre cost*. A family designed for easy torsion or isogeny descent can therefore have insufficient capacity before any search is run.

2.2 Specialization and visibility

A fixed independent generic subgroup specializes injectively away from a controlled exceptional set, but extra points discovered after specialization can be dependent. Given a generic lattice with Gram matrix G_0 and candidate extra points with block matrix

$$G = \begin{pmatrix} G_0 & B \\ B^T & C \end{pmatrix},$$

the new directions are measured by the Schur complement

$$S = C - B^T G_0^{-1} B.$$

The relevant search height is the norm in the quotient lattice represented by S , not merely the absolute canonical height of a candidate point.

In a non-isotrivial family, specialized canonical heights grow with the height of the base parameter. A curve can therefore have genuinely high rank while its missing generators lie beyond a feasible coordinate search. This is a visibility obstruction, not a rank obstruction.

2.3 Descent and the local–global gap

A large 2- or 3-Selmer group records locally soluble covering classes. The excess over visible Mordell–Weil rank may lie in Sha. Consequently,

$$\text{large Selmer dimension} \not\Rightarrow \text{many rational points.}$$

The correct construction problem is to make the relevant covering curves globally soluble and of manageable height, preferably through shared low-genus geometry.

2.4 Auxiliary-base arithmetic

Extra points on fibres are naturally represented by multisections or covering curves. A genus-zero auxiliary curve with a rational point is parametrizable. A genus-one auxiliary curve is useful when its Jacobian has positive rational rank. A higher-genus auxiliary curve can have only finitely many rational points, although finiteness never implies emptiness.

Thus the useful resource is not only generic rank. It is

forced rank + low auxiliary genus + positive-rank parameter space + low quotient height.

2.5 Certification cost

A lower-bound certificate does not require complete saturation of the full Mordell–Weil group, but it does require exact independence of the displayed points. Finite-field reductions, interval-certified heights, and a second computer algebra system should be used incrementally. Search code and verification code are deliberately separated so that a verifier depends only on the equation and point list.

2.6 What the historical graph does not prove

Record dates are not independent observations from a stationary process. Each jump reflects a different family, fibration, sieve, point-search method, amount of computation, and publication decision. A fitted “inflection point” is therefore not evidence of a universal rank ceiling. Failure of a family, a bounded search, or a low-height enumeration yields only a restricted obstruction unless it is converted into a theorem.

3 Galois-character packets and exact failed routes

3.1 Multiquadratic branch codes

Let $K = \mathbf{Q}(t)$, let $D \subset K^\times/K^{\times 2}$ be a k -dimensional space of independent geometric squareclasses, and put

$$L_D = K(\sqrt{d} : d \in D).$$

Let C_D be the smooth projective curve with function field L_D , and let b be the total number of geometric branch points.

The branch-parity vector of a rational function has even weight. Since $\mathbf{P}_{\mathbf{Q}}^1$ has no nontrivial unramified double cover, the branch map embeds D into the even-weight code on b coordinates. Hence

$$k \leq b - 1.$$

Every branch point has inertia group of order two. Riemann–Hurwitz gives, for $k \geq 2$,

$$g(C_D) = 1 + 2^{k-2}(b - 4).$$

Therefore

$$g(C_D) \geq 1 + 2^{k-2}(k-3).$$

Thirteen independent quadratic characters require at least fourteen branch points and hence genus at least

$$1 + 2^{11}(14-4) = 20481.$$

Low genus is already lost at the fourth independent character.

3.2 The complete twist packet

After tensoring with $\mathbf{Q}$, character idempotents give the exact decomposition

$$E(L_D) \otimes \mathbf{Q} \cong \bigoplus_{d \in D} E^d(K) \otimes \mathbf{Q}.$$

Distinct character summands are orthogonal under the canonical height. In the biquadratic case,

$$\text{rank } E(K(\sqrt{d_1}, \sqrt{d_2})) = \text{rank } E(K) + \text{rank } E^{d_1}(K) + \text{rank } E^{d_2}(K) + \text{rank } E^{d_1 d_2}(K).$$

The product-twist channel is often omitted when a search records only the two multisections used to define the cover.

A single character can carry more than one direction. The repository contains an exact family where one nontrivial quadratic eigenspace has rank at least three. Thus the branch-code theorem bounds the number of low-genus characters, not the total rank of their Mordell–Weil packets.

3.3 A long and useful failed route

The first proposed mechanism started with a rank-17 K3 fibration, constructed thirteen quadratic multisections, and attempted to split all thirteen over one low-genus base. It seemed capable of forcing $17 + 13 = 30$.

The genus formula proves that the intended low-genus family cannot arise from thirteen independent squareclasses. This is the smallest decisive obstruction: the fourth independent character already forces genus at least five.

3.4 What the failure reveals

The useful surviving data are:

- the squareclass of every multisection;
- all sections in the same character channel;
- every product-twist channel;
- the exact Shioda Gram matrix within each channel;
- branch support, auxiliary genus, and global-solubility data.

The search object is therefore a complete *Galois-character height packet*, not a list of individually constructed points.

3.5 One-function visibility barrier

Let f be a rational function on an elliptic curve with divisor

$$(f) = P_1 + \cdots + P_N - NO.$$

Since a principal divisor maps to the identity in $\text{Pic}^0(E)$,

$$\boxed{P_1 + \cdots + P_N = O.}$$

Consequently, a construction producing exactly thirty rational points as the zeros of one function has a forced integral dependence and can span rank at most twenty-nine.

For a short Weierstrass equation $y^2 = R(x)$ and a function $f = A(x) + yB(x)$, the norm is

$$\text{Norm}(f) = A(x)^2 - R(x)B(x)^2.$$

The corrected one-function target is therefore a split degree-31 identity

$$A(x)^2 - R(x)B(x)^2 = c \prod_{i=1}^{31} (x - r_i), \quad r_i \in \mathbf{Q}.$$

The thirty-one points have one unavoidable relation, leaving room for rank thirty. This remains an independent exact construction track.

4 The split E_8 rational surface

4.1 Why torsion-friendly models are too small

A rational elliptic surface with rational two-torsion can be written

$$y^2 = x^3 + A(t)x^2 + B(t)x, \quad \deg A \leq 2, \quad \deg B \leq 4.$$

Its discriminant is

$$\Delta = 16B^2(A^2 - 4B).$$

The zeros of B force reducible fibres of total root rank at least four, so Shioda–Tate bounds the generic Mordell–Weil rank by four. A smooth three-character V_4 packet built from such bases has total capacity at most twenty-two.

Rational three-torsion in Tate normal form forces still more reducible-fibre cost; the analogous packet capacity is at most fourteen. The easiest descent models are therefore incapable of carrying rank thirty. The high-capacity base should have trivial small rational torsion and twelve irreducible I_1 fibres.

4.2 A certified split E_8 surface

The repository contains an exact certificate for the rational elliptic surface

$$\mathcal{E} : \quad y^2 = x^3 + A(t)x + B(t),$$

with

$$\begin{aligned} A(t) &= -27t^4 + 216t^3 + 6t^2 - 432t - 267, \\ B(t) &= 54t^6 + 108t^5 - 630t^4 - 880t^3 + 1458t^2 + 1278t + 1242. \end{aligned}$$

Its discriminant has twelve simple roots, so every singular fibre is of type I_1 . The surface is rational and has no reducible-fibre root lattice. Eight explicit rational sections are stored in

the machine-readable certificate. Their Shioda height matrix is even, positive definite, has rank eight, and determinant one. Therefore the generic Mordell–Weil lattice is exactly

$$\boxed{E_8.}$$

This is the preferred base for packet engineering: it realizes the maximum rank of a rational elliptic surface without spending Picard rank on reducible fibres.

4.3 Why higher- χ neighbour fibrations do not replace K3 methods

Let $S \rightarrow \mathbf{P}^1$ be a relatively minimal elliptic surface with section and $d = \chi(\mathcal{O}_S)$. The canonical bundle formula gives

$$K_S \sim (d - 2)F.$$

If $d \neq 2$, the canonical class determines the primitive fibre ray. Any other genus-one fibre class F' satisfies $K_S \cdot F' = 0$, hence $F \cdot F' = 0$; the Hodge index theorem forces F' to be proportional to F . Primitivity gives $F' = F$.

Thus

$$\boxed{d \neq 2 \implies \text{the genus-one fibration is canonically rigid.}}$$

K3 surfaces, where $d = 2$ and $K_S = 0$, are the exceptional case in which neighbour fibrations can genuinely move lattice directions between fibre components and sections. This explains why changing the K3 fibration broke the rank-28 plateau, while the same manoeuvre cannot simply be transplanted to an arithmetic-genus-four surface.

5 The minimal cubic genus-one mechanism

5.1 Low-genus pullback capacity

Let $\pi : S \rightarrow C$ be a relatively minimal non-isotrivial elliptic surface with section, and put

$$\chi = \chi(\mathcal{O}_S), \quad g = g(C).$$

The standard surface identities give

$$e(S) = 12\chi, \quad q(S) = g, \quad p_g(S) = \chi + g - 1.$$

Consequently

$$b_2(S) = 12\chi - 2 + 4g, \quad \boxed{h^{1,1}(S) = 10\chi + 2g.}$$

If R is the total root rank of reducible fibres, Shioda–Tate yields

$$\boxed{\text{rank } E(\overline{\mathbf{Q}}(C)) \leq 10\chi + 2g - 2 - R.}$$

After a degree- d base change of a rational elliptic surface, $\chi = d$. Hence

$$\text{rank } E(\overline{\mathbf{Q}}(C)) \leq 10d + 2g(C) - 2 - R.$$

For a base with infinitely many rational points, the smallest possibilities capable of rank thirty are

$$\boxed{(d, g) = (4, 0) \quad \text{or} \quad (d, g) = (3, 1).}$$

The second possibility is the decisive correction to the earlier assumption that degree four was minimal.

5.2 Ramification must avoid the discriminant

Suppose the rational elliptic surface has twelve I_1 fibres. Ramification of index e over an I_1 fibre produces an I_e fibre and consumes root rank $e - 1$. A degree-three genus-one cover has total ramification six by Riemann–Hurwitz. A cyclic cubic cover realizes this through three totally ramified points.

Therefore a cubic pullback can reach the ceiling thirty only when all three branch points lie over smooth fibres. Branching over even one I_1 fibre lowers the Shioda–Tate ceiling below thirty.

5.3 An explicit positive-rank cyclic base

Consider

$$C_{36} : \quad u^3 = 36t(t-1).$$

The map $(t, u) \mapsto t$ is a cyclic cubic cover after adjoining a primitive cube root of unity, totally ramified over $0, 1, \infty$. It is a genus-one curve. More generally, for

$$u^3 = ct(t-1),$$

put

$$X = 4cu, \quad Y = 4c^2(2t-1).$$

Then

$$Y^2 = X^3 + 16c^4.$$

For $c = 36$, the rational point $(t, u) = (3, 6)$ maps to

$$P = (864, 25920).$$

Doubling gives

$$x(2P) = \frac{3456}{25},$$

which is nonintegral. If P were torsion then $2P$ would be torsion, in contradiction with Lutz–Nagell on the integral Mordell model. Hence

$$\boxed{\text{rank } C_{36}(\mathbf{Q}) \geq 1.}$$

A Möbius transformation moves the three branch values to any chosen triple of rational smooth fibres.

5.4 Cyclic cubic eigenspaces

Let $C \rightarrow \mathbf{P}^1$ be such a cyclic cubic cover and let σ generate its deck group. Over $\mathbf{Q}(\zeta_3)$, the Mordell–Weil space of the pullback decomposes as

$$M = M_1 \oplus M_\zeta \oplus M_{\zeta^2}.$$

The invariant part is the pullback of the original E_8 lattice, so

$$\dim M_1 = 8.$$

Complex conjugation exchanges the two nontrivial character spaces. Writing $\dim M_\zeta = \dim M_{\zeta^2} = r$, one obtains

$$\boxed{\text{rank } M = 8 + 2r.}$$

The Hodge bound gives $r \leq 11$. In the absence of reducible fibres,

$$\boxed{\text{rank } M = 30 \iff r = 11 \iff \rho(S_C) = 32.}$$

The arithmetic problem is not merely Picard maximality: the thirty geometric directions must descend to $\mathbf{Q}(C)$.

6 The order-three Hodge representation

The number eleven in the cyclic cubic target is forced by the Hodge representation; it is not merely the difference between thirty and the base rank eight.

Let S_C be the cubic genus-one pullback. It has

$$\chi(\mathcal{O}_{S_C}) = 3, \quad q = 1, \quad p_g = 3, \quad e = 36.$$

Hence

$$b_2 = 38, \quad h^{1,1} = 32.$$

Let σ generate the deck group. Its fixed locus is the union of the three smooth elliptic fibres above the branch points, whose Euler number is zero. On H^1 and H^3 , the trace is -1 , because the quotient of the base curve is $\mathbf{P}^1$. The topological Lefschetz formula gives

$$\mathrm{Tr}(\sigma|H^2) = -4.$$

Writing the dimensions of the three complex character spaces as

$$n_0, n, n,$$

one has

$$n_0 + 2n = 38, \quad n_0 - n = -4.$$

Therefore

$$(n_0, n, n) = (10, 14, 14).$$

The holomorphic two-forms have character multiplicities

$$(0, 2, 1)$$

up to exchanging the two nontrivial characters. Complex conjugation reverses the last two entries for $H^{0,2}$. It follows that

$$\dim H_1^{1,1} = 10, \quad \dim H_{\zeta_3}^{1,1} = \dim H_{\zeta_3^2}^{1,1} = 11.$$

Since all singular fibres remain irreducible,

$$\mathrm{rank} \mathrm{MW}(S_C) = \rho(S_C) - 2.$$

The invariant Mordell–Weil lattice is $E_8(3)$, and each nontrivial character has capacity eleven. Thus the following are equivalent:

$$\begin{aligned} &\mathrm{rank} \mathrm{MW}(S_C) = 30, \\ &\rho(S_C) = 32 = h^{1,1}(S_C), \\ &\dim \mathrm{MW}_{\zeta_3} = \dim \mathrm{MW}_{\zeta_3^2} = 11, \\ &\text{all nontrivial-character } (1, 1) \text{ classes are algebraic.} \end{aligned}$$

A rank-thirty cubic pullback is therefore a Picard-maximal irregular elliptic surface with a trace-zero Eisenstein Mordell–Weil lattice of complex rank eleven. This connects the explicit trisection search to order-three Noether–Lefschetz loci, CM points, Delsarte surfaces, and ternary lattice gluing.

7 Cyclic trisections and the ternary trace code

7.1 Trisection shells in E_8

Let S be the split E_8 rational elliptic surface, with zero section O and fibre class F . Its Néron–Severi lattice decomposes as

$$\mathrm{NS}(S) = \langle O, F \rangle \oplus E_8(-1), \quad O^2 = -1, \quad O \cdot F = 1, \quad F^2 = 0.$$

Let D be a genus-one trisection and put

$$m = D \cdot O \geq 0.$$

Writing

$$D = 3O + (m+3)F + v, \quad v \in E_8(-1),$$

and using $K_S = -F$, adjunction gives $D^2 = 3$. Therefore

$$\boxed{\|v\|_{E_8}^2 = 6(m+1)}.$$

The theta series

$$\Theta_{E_8}(q) = 1 + 240 \sum_{n \geq 1} \sigma_3(n) q^n$$

shows that the number of such lattice classes is

$$\boxed{240\sigma_3(3(m+1))}.$$

For $m = 0$ this gives exactly 6720 minimal trisection classes.

7.2 Universal projected orbit geometry

Assume the normalization of D is a cyclic cubic genus-one cover with simple total ramification over three smooth fibres. On the pullback, let

$$P_0 = P, \quad P_1 = \sigma P, \quad P_2 = \sigma^2 P.$$

Distinct conjugates meet transversely at the three ramification points and nowhere else, so

$$P_i \cdot P_j = 3 \quad (i \neq j).$$

The pulled-back surface has $\chi = 3$ and no reducible fibres. Shioda’s formula yields

$$\langle P_i, P_i \rangle = 6 + 2m, \quad \langle P_i, P_j \rangle = 2m \quad (i \neq j).$$

Let

$$T = P_0 + P_1 + P_2, \quad q_i = P_i - \frac{T}{3}.$$

Then

$$\langle T, T \rangle = 18(m+1)$$

and, independently of m ,

$$\boxed{\langle q_i, q_i \rangle = 4, \quad \langle q_i, q_j \rangle = -2 \quad (i \neq j)}.$$

Thus every cyclic trisection orbit contributes a projected plane

$$A_2(2), \quad \mathrm{Gram} = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}.$$

The integral difference lattice generated by $P_0 - P_1$ and $P_1 - P_2$ is

$$A_2(6), \quad \text{Gram} = \begin{pmatrix} 12 & -6 \\ -6 & 12 \end{pmatrix}.$$

This corrects an attractive but impossible earlier fingerprint. Because the surface has $\chi = 3$ and no reducible fibres, every nonzero section has height at least six. The full Mordell–Weil lattice cannot contain roots of norm two or four; norm four appears only after orthogonal projection away from the invariant $E_8(3)$ lattice.

7.3 The mod-3 trace code

The trace T is the pullback of the E_8 vector v . For two orbit sections P, Q with traces v, w , Galois invariance gives

$$\langle q_P, q_Q \rangle = \langle P, Q \rangle - \frac{\langle v, w \rangle_{E_8}}{3}.$$

Consequently, an integral projected packet requires the trace residues to be mutually orthogonal in $E_8/3E_8$. Every trisection trace is isotropic modulo three because its norm is divisible by six.

The finite quadratic space $E_8/3E_8$ is split of dimension eight and Witt index four. Exact enumeration gives:

- 2240 nonzero isotropic vectors;
- exactly three norm-six E_8 lifts of each nonzero isotropic residue;
- 2240 maximal totally isotropic four-spaces;
- 80 nonzero residues and hence 240 minimal trisection classes in each maximal four-space;
- 80 maximal four-spaces through each nonzero isotropic residue.

The projective-quadric counts are

$$(3^3 + 1)(3^4 - 1) = 2240$$

for nonzero isotropic vectors and

$$2 \prod_{i=1}^3 (3^i + 1) = 2240$$

for maximal isotropic four-spaces. The three-to-one lifting statement is verified by the dependency-free exact enumeration shipped with the paper.

7.4 Determinant identity

Let M be the full Mordell–Weil lattice of the cubic pullback and let

$$L = M^{(\sigma)} \cong E_8(3).$$

Let K be the image of M under orthogonal projection to the trace-zero rational subspace. If K is integral, the projection exact sequence and an orthogonal covolume calculation give

$$\det M = \det L \det K = 3^8 \det K.$$

This determinant is a powerful adversarial check: eleven apparent orbit planes must be tested not only for rational independence but also for ternary gluing and saturation.

7.5 Finite search reduction

The minimal integral track is no longer an undifferentiated search through all 6720 norm-six classes. It is the finite incidence problem

$$\boxed{2240 \text{ trace codes} \times 240 \text{ minimal classes per code.}}$$

For each code, one computes the four-dimensional systems $|D_v|$, retains members whose cubic discriminant is a square and whose branch divisor consists of three smooth fibres, canonicalizes the resulting covers, and groups all trisections defining the same cover. Eleven Hermitian-independent orbit planes over one positive-rank cover would contribute twenty-two new directions; together with $E_8(3)$ they would force rank thirty.

8 Marked cubic surfaces and outer Galois points

The trace-code theorem becomes substantially more concrete in the minimal norm-six shell.

8.1 Unique decomposition of a norm-six trace

Every $v \in E_8$ with $v^2 = 6$ has a unique unordered decomposition

$$\boxed{v = \alpha + \beta}$$

where α and β are roots with

$$\alpha^2 = \beta^2 = 2, \quad \langle \alpha, \beta \rangle = 1.$$

Indeed, a root α occurs precisely when $\langle v, \alpha \rangle = 3$, for then $v - \alpha$ is again a root. There can be at most two such roots, and they are complementary. Counting gives the same result: each of the 240 roots has 56 roots of inner product one, so the number of unordered pairs is

$$\frac{240 \cdot 56}{2} = 6720,$$

exactly the size of the norm-six shell.

Let S_α, S_β be the corresponding root sections. They are pairwise disjoint from each other and from O , because

$$S_\alpha \cdot S_\beta = 1 - \langle \alpha, \beta \rangle = 0.$$

The minimal trisection divisor satisfies

$$\boxed{D_v = F + O + S_\alpha + S_\beta.}$$

8.2 Anticanonical cubic-surface model

Contract the three disjoint (-1) -curves

$$O, \quad S_\alpha, \quad S_\beta$$

to obtain $\varphi : S \rightarrow X$. Since $K_S = -F$,

$$K_S = \varphi^* K_X + O + S_\alpha + S_\beta$$

and hence

$$\boxed{D_v = \varphi^*(-K_X).}$$

The surface X is a smooth degree-three del Pezzo surface. Its anticanonical system embeds it as a smooth cubic surface in $\mathbf{P}^3$, and $|D_v|$ is the pullback of the hyperplane system.

Let p_0, p_1, p_2 be the images of the three contracted sections. The elliptic pencil $|F|$ is the strict transform of anticanonical hyperplanes through all three points. Since this system is one-dimensional, the points are collinear. Their line ℓ is not contained in X , and

$$\ell \cap X = p_0 + p_1 + p_2.$$

For a hyperplane H not containing ℓ , the curve

$$C_H = X \cap H$$

is a plane cubic and the induced trisection map is projection from

$$q_H = H \cap \ell.$$

Thus the Riemann–Roch search is equivalent to a marked cubic-surface problem.

8.3 Cyclicity is an outer Galois-point condition

The degree-three projection $C_H \rightarrow \mathbf{P}^1$ is cyclic exactly when q_H is an outer Galois point of the plane cubic. Over an algebraic closure, one can then choose coordinates with

$$q_H = (1 : 0 : 0), \quad C_H : \quad X^3 + G_3(Y, Z) = 0.$$

The deck transformation is $X \mapsto \zeta_3 X$. In particular,

$$\boxed{j(C_H) = 0.}$$

Conversely, the displayed outer-Galois form gives a cyclic cubic trisection. Descent to $\mathbf{Q}$ and positive rational rank remain separate arithmetic tests.

The new exact search is therefore:

1. recover the unique root pair for each trace class;
2. construct the marked cubic surface (X, ℓ) ;
3. solve for hyperplanes whose moving projection centre is an outer Galois point;
4. retain smooth $j = 0$ curves with positive rational rank;
5. group distinct trace classes producing the same cover and compute their Hermitian rank.

The first decisive milestone is one common cyclic cover arising from two distinct trace classes. The final target is eleven Hermitian-independent classes over one positive-rank cover.

9 Hessian reduction of the cyclic locus

Let $X = \{F = 0\} \subset \mathbf{P}^3$ be the marked cubic surface attached to a minimal trace class, and let ℓ be the 3-secant line through the three contracted section points. A hyperplane H not containing ℓ determines

$$q = H \cap \ell, \quad C_H = X \cap H,$$

and the trisection map is projection of the plane cubic C_H from q .

9.1 The invariant polar form

Write $F(v) = T(v, v, v)$ for the symmetric trilinear polarization. For $q \notin X$, put

$$a = F(q), \quad \lambda_q(u) = T(q, q, u),$$

and define

$$B_q(u, v) = aT(q, u, v) - \lambda_q(u)\lambda_q(v).$$

This form descends to the three-dimensional quotient $V/\langle q \rangle$. If $H = \langle q, W \rangle$, then in coordinates on H with $q = (1 : 0 : 0)$ the restricted cubic is

$$F_H = ax^3 + x^2L(y, z) + xQ(y, z) + C(y, z).$$

The substitution

$$x = X - \frac{L}{3a}$$

removes the X^2 term. Projection is cyclic precisely when the remaining mixed quadratic vanishes:

$$3aQ - L^2 = 0.$$

In invariant form,

$$B_q|_W = 0.$$

9.2 The Hessian criterion

In a basis beginning with q , write

$$M_q = \begin{pmatrix} a & \lambda^T \\ \lambda & Q_q \end{pmatrix}$$

for the matrix of $T(q, -, -)$. The matrix of B_q on the quotient is $aQ_q - \lambda\lambda^T$. The block determinant formula gives

$$\det B_q = a^2 \det M_q.$$

Since the Hessian matrix of F at q is $6M_q$,

$$\det B_q = 0 \iff q \in \text{Hess}(X).$$

Over an algebraic closure, a symmetric form on a three-dimensional space has a two-dimensional totally isotropic subspace exactly when it is singular. Therefore a geometrically cyclic hyperplane section is equivalent to a point $q \in \ell \cap \text{Hess}(X)$, outside X , together with a maximal isotropic plane of B_q .

9.3 A finite quartic search

The Hessian of a cubic surface is quartic. If ℓ is not contained in it, there are at most four possible projection centres. At a generic rank-two Hessian point, the form B_q has exactly two maximal isotropic planes over the algebraic closure. Thus a generic trace class has at most

$$8$$

geometrically cyclic hyperplane sections.

For a maximal ternary trace code with 240 minimal classes, the continuous hyperplane search collapses to at most

$$\boxed{240 \cdot 8 = 1920}$$

explicit candidates. Each is then tested for rational descent, smoothness, avoidance of the elliptic discriminant, and positive rational rank.

This is the current decisive representation. The search is no longer an unstructured square-discriminant condition in a projective three-space. It is one Hessian quartic restricted to one 3-secant line, followed by the solution of a rank-two quadratic form.

10 Exact Eisenstein packet determinants

For orbit representatives $P_1, \dots, P_n$, put

$$h_{ij}^{(k)} = \langle P_i, \sigma^k P_j \rangle, \quad k = 0, 1, 2,$$

and define the Eisenstein Hermitian matrix

$$\boxed{\mathcal{H}_{ij} = h_{ij}^{(0)} + \zeta_3 h_{ij}^{(1)} + \zeta_3^2 h_{ij}^{(2)}}.$$

Galois invariance makes $\mathcal{H}$ Hermitian over $\mathbf{Z}[\zeta_3]$. Every diagonal entry equals six, independently of the intersection of the trisection with the zero section.

Let $T_i = P_i + \sigma P_i + \sigma^2 P_i$ have E_8 trace vector v_i . Then

$$h_{ij}^{(0)} + h_{ij}^{(1)} + h_{ij}^{(2)} = \langle v_i, v_j \rangle_{E_8}.$$

The projected pairings are integral precisely when the trace residues obey the mod-three orthogonality constraints.

The difference lattice generated by

$$P_i - \sigma P_i, \quad \sigma P_i - \sigma^2 P_i$$

is the trace form of the Hermitian lattice $\mathcal{H}$. Since $|\text{disc } \mathbf{Q}(\zeta_3)| = 3$,

$$\boxed{\det L_{\text{diff}} = 3^n \text{Norm}(\det \mathcal{H})}.$$

Multiplication by $1 - \zeta_3$ maps the projected lattice onto the difference lattice with index 3^n . Therefore

$$\boxed{\det L_{\text{proj}} = 3^{-n} \text{Norm}(\det \mathcal{H})}.$$

Including the invariant $E_8(3)$ lattice gives

$$\boxed{\det L_{\text{packet}} = 3^{8-n} \text{Norm}(\det \mathcal{H})}.$$

For the rank-thirty target $n = 11$,

$$\boxed{\det L_{30} = \frac{\text{Norm}(\det \mathcal{H})}{27}}.$$

This formula is an exact early filter. A putative eleven-orbit packet must have nonzero Hermitian determinant, satisfy the ternary integrality conditions, have norm divisible by twenty-seven, and yield a determinant compatible with the integral Néron–Severi lattice and saturation. The full thirty-by-thirty real Gram matrix is needed only after these cheaper exact checks pass.

11 A minimal-character scale obstruction in the marked $j = 0$ packet

The cyclic cubic strategy admits a particularly sharp exact test in the marked $j = 0$ family. Put

$$q(t) = t^2 - t + 1, \quad r(t) = t(t - 1),$$

and specialize the marking parameter to $\mu = 2$, so that

$$a = q(2) = 3, \quad b = \frac{q(2)^3}{r(2)} = \frac{27}{2}.$$

After the cyclic base change

$$u^3 = ct(t - 1), \quad c \in \mathbf{Q}^*,$$

the surface over $K = \mathbf{Q}(\sqrt{-3})$ is

$$y^2 = x^3 + \frac{b^2}{c^2}u^6 - a^3 \left(\frac{u^3}{c} + 1 \right)^3.$$

Theorem 11.1 (Minimal-character scale obstruction). *A character-one section in the minimal polynomial box*

$$x = u(Au^3 + B), \quad y = Cu^6 + Du^3 + E, \quad A, B, C, D, E \in K,$$

can exist only when

$$[c] \in \{[1], [2]\} \subset \mathbf{Q}^*/\mathbf{Q}^{*3}.$$

Both classes occur and carry a non-torsion Eisenstein eigenline over K . However, the corresponding genus-one bases have

$$\text{rank } C_1(\mathbf{Q}) = \text{rank } C_2(\mathbf{Q}) = 0.$$

Thus no positive-rank cyclic base in this $\mu = 2$ family carries a minimal character-one eigensection.

Proof. Coefficient comparison gives

$$\begin{aligned} C^2 &= A^3, & 2CD &= 3A^2B - \frac{27}{c^3}, \\ D^2 + 2CE &= 3AB^2 + \frac{405}{4c^2}, & 2DE &= B^3 - \frac{81}{c}, \\ E^2 &= -27. \end{aligned}$$

The branch $B = 0$ forces $D = 3E/(2c)$ and $C = E/(3c^2)$, whence $D^2 + 2CE = -315/(4c^2)$, a contradiction. Hence $B \neq 0$. Choose $E = 3\sqrt{-3}$ and set

$$\rho = \frac{cA}{B}, \quad S = cB^3, \quad g = \frac{c^2C}{\sqrt{-3}}, \quad d = \frac{cD}{\sqrt{-3}}.$$

Exact lexicographic elimination gives

$$(2\rho + 1)^3(10\rho^3 - 48\rho^2 + 12\rho - 1) = 0.$$

The cubic factor is irreducible over $\mathbf{Q}$, so it has no root in the quadratic field K . Therefore $\rho = -1/2$. The two remaining normalized solutions are

$$(S, g, d) = \left(54, -\frac{3}{2}, \frac{3}{2}\right), \quad (S, g, d) = \left(216, 3, -\frac{15}{2}\right).$$

If $z \in K$ and $z^3 \in \mathbf{Q}$, then $z^3 \in \mathbf{Q}^3$; hence $K^3 \cap \mathbf{Q} = \mathbf{Q}^3$. Since $S/c = B^3$, the class of c equals the class of S . Now $54 = 2 \cdot 3^3$ and $216 = 6^3$, proving the two-class statement. Exact substitution supplies sections for both classes. Their specializations at $u = 1$ have reduction orders $(13, 19)$ at primes above $(7, 13)$ in class [1], and $(13, 21)$ in class [2]. The good-reduction constraints on a hypothetical torsion order are incompatible, so the sections are non-torsion. Finally, SageMath with proof flags computes exact rank zero for the two base curves, which are birational to $Y^2 = X^3 + 16c^4$. $\square$

This theorem explains an earlier computational split. The scale $c = 27$ is cube-equivalent to $c = 1$ and exhibits minimal eigensections, but its base has rank zero. The scale $c = 36$ has base rank one, but the complete minimal scheme is empty. The incompatibility is structural for this marked surface, not a failure of a small coefficient box.

11.1 Decisive change of setting

The minimal ansatz is exhausted. The next exact experiment is a controlled- pole character-one search on the rank-one base C_3 , followed in parallel by the same Kummer classification along the CM-compatible marking conic $q(\mu) = 3s^2$. A successful one-pole orbit would contribute a genuine Eisenstein line over a positive-rank genus-one base; failure would give a second, sharper visibility obstruction.

12 The hidden quadratic twist and a rank-twelve ceiling

The minimal-scale obstruction is a symptom of a stronger geometric failure. For every nonzero rational c , the complete marked $\mu = 2$ cubic-base family has geometric generic rank at most twelve.

Put $v = t(t - 1) = u^3/c$. The coefficient factors as

$$\frac{729}{4}v^2 - 27(v + 1)^3 = -\frac{27}{4}(v - 2)^2(4v + 1).$$

Since t satisfies

$$t^2 - t - \frac{u^3}{c} = 0,$$

the function field of the genus-one base is

$$\mathbf{Q}(C_c) = \mathbf{Q}\left(u, \sqrt{1 + \frac{4u^3}{c}}\right).$$

The Mordell–Weil space therefore decomposes into the invariant curve over $\mathbf{Q}(u)$ and its quadratic twist by

$$D_c(u) = 1 + \frac{4u^3}{c}.$$

Theorem 12.1 (Rank-twelve ceiling for the marked cubic family). *For every $c \in \mathbf{Q}^*$,*

$$\text{rank } E(\overline{\mathbf{Q}}(C_c)) \leq 12.$$

Consequently a rank-thirty specialization inside this family would require an exceptional rank jump of at least eighteen.

Proof. The invariant surface has coefficient

$$-\frac{27}{4} \left(\frac{u^3}{c} - 2 \right)^2 \left(\frac{4u^3}{c} + 1 \right).$$

It is a K3 surface with three fibres of type IV , three of type II , and an I_0^* fibre at infinity. The total root rank is

$$3 \cdot 2 + 4 = 10,$$

so Shioda–Tate and $\rho \leq h^{1,1} = 20$ give invariant rank at most eight. The Euler number check is

$$3 \cdot 4 + 3 \cdot 2 + 6 = 24.$$

For a short $j = 0$ model, quadratic twisting multiplies a_6 by D_c^3 . The twist coefficient is therefore

$$-\frac{27}{4} \left(\frac{u^3}{c} - 2 \right)^2 \left(\frac{4u^3}{c} + 1 \right)^4.$$

This is a $\chi = 3$ elliptic surface with three fibres of type IV and three of type IV^* . Its root rank is

$$3 \cdot 2 + 3 \cdot 6 = 24.$$

Since $h^{1,1} = 30$, its Mordell–Weil rank is at most

$$30 - 2 - 24 = 4.$$

Its Euler number is $3 \cdot 4 + 3 \cdot 8 = 36$. The quadratic-extension rank decomposition now gives the bound $8 + 4 = 12$. $\square$

This calculation shows why the ambient Hodge capacity of a degree-three elliptic base is not enough: repeated factors create reducible fibres whose root lattices consume eighteen of the twenty-two hoped-for trace-zero directions. The $\mu = 2$ marked family should therefore no longer be treated as a primary generic rank-thirty candidate.

12.1 New setting

The cyclic-cubic programme now moves along the CM-compatible marking conic $q(\mu) = 3s^2$ and applies the hidden-quadratic fibre-cost test before any section search. Only parameters whose invariant and twist eigenspaces retain combined capacity at least thirty are promoted to the expensive trisection incidence stage.

13 The six-channel capacity criterion

The degeneration at $\mu = 2$ does not occur for a generic rational marking. Let

$$a = \mu^2 - \mu + 1, \quad b = \frac{a^3}{\mu(\mu - 1)}, \quad P_\mu(v) = b^2 v^2 - a^3(v + 1)^3.$$

Exact factorization gives

$$P_\mu(v) = -\frac{a^3}{\mu^2(\mu - 1)^2} (v - \mu(\mu - 1))(\mu^2 v + \mu - 1)((\mu - 1)^2 v - \mu).$$

Thus the three roots are rational. They are distinct and avoid 0 and $-1/4$ outside

$$\mu \in \left\{ 0, 1, \frac{1}{2}, 2, -1 \right\}.$$

Over an algebraic closure, the full cyclic cubic genus-one base is obtained by adjoining $u^3 = cv$ and $\sqrt{1+4v}$. The $C_3 \times C_2$ character decomposition consists of the six sextic-twist channels

$$E_{i,\varepsilon} : \quad y^2 = x^3 + v^{2i}(1+4v)^{3\varepsilon}P_\mu(v), \quad i = 0, 1, 2, \quad \varepsilon = 0, 1.$$

Theorem 13.1 (Six-channel capacity). *Away from the five degenerate markings, the geometric Mordell–Weil rank capacities of the six channels are*

$$\boxed{(4, 6, 4, 4, 6, 6)}.$$

Their sum is exactly thirty. The three rational elliptic surface channels have exact geometric ranks 4, 6, 4. Consequently the full packet has geometric rank thirty if and only if the three K3 channels have Picard number twenty, equivalently if they attain ranks

$$\boxed{4, 6, 6}.$$

Proof. The roots of P_μ , 0, $-1/4$, and infinity are distinct. Applying the Kodaira table to the exponents of $v^{2i}(1+4v)^{3\varepsilon}P_\mu(v)$ gives the following root ranks:

$$4, 2, 14, 4, 12, 12.$$

The corresponding arithmetic genera are

$$1, 1, 2, 1, 2, 2.$$

Since $h^{1,1} = 10\chi$ over $\mathbf{P}^1$, Shioda–Tate gives the capacity vector $(4, 6, 4, 4, 6, 6)$. Rational elliptic surfaces have Picard number ten, so the three $\chi = 1$ bounds are attained automatically. The remaining sixteen directions are attained precisely when each K3 surface has Picard number twenty. $\square$

This is the first exact reduction of the rank-thirty target to three finite Picard-maximality tests. The scale c affects rational descent and the rank of the genus-one base but not the geometric capacity vector.

13.1 Finite-field rejection sieve

For each rational marking, the three K3 channels are reduced at several good primes and their exact finite-function-field geometric ranks are computed. Any rank below $(4, 6, 6)$ rejects the marking. A survivor is not a proof of characteristic-zero rank: it is promoted only after explicit sections and an independent height certificate are constructed.

14 Lean verification of the exact algebraic layer

The repository contains a machine-checked Lean 4 formalization, pinned to Lean and mathlib version 4.32.2. The root module is `EllipticRank30.lean`; continuous integration builds it and rejects local `sorry`, `admit`, and `axiom` declarations.

The Lean development certifies the following exact ingredients used in this paper:

- the birational identity taking $u^3 = ct(t-1)$ to $Y^2 = X^3 + 16c^4$;
- positivity of $\mu^2 - \mu + 1$ over the reals and the resulting impossibility of the minimal real constant equation;

- the generic marked-coefficient factorization and the symmetric product and pairwise-product identities for its three rational roots;
- the repeated-factor and quadratic-twist identities at the degenerate marking $\mu = 2$;
- both normalized rational solutions used in the minimal-scale obstruction;
- every numerical Euler, root-rank, rank-capacity, shell-size, and trace-code count used in the six-channel reduction.

This formalization has a deliberately narrow truth boundary. It does not yet formalize Kodaira classification, minimal elliptic surfaces, Shioda–Tate, Picard-number bounds, character decomposition of Mordell–Weil groups, function-field rank computations, canonical heights, saturation, or rational point independence. Those statements remain supported by the exact proofs and independent SageMath/Magma certificates described elsewhere in the paper. Thus a green Lean build strengthens the algebraic core but is not, by itself, a rank-thirty certificate.

15 Executable search program

The search is organized around proof objects rather than scores. Every candidate record stores exact coefficients, rational points, local data, height information, subgroup rank, regulator, saturation status, software versions, provenance, and a content hash.

15.1 Workstream A: the second rank-17 K3 fibration

The successful rank-29 specialization arose from a second rank-17 fibration on the record K3 surface. The ideal reproduction input consists of its exact $\mathbf{Q}(t)$ Weierstrass model, all seventeen labelled sections, their Shioda Gram matrix, the bad-parameter locus, and the coordinate transformation to the published record fibre. Until those data are available in executable form, the project does not replace them with a guessed family.

Once ingested, the search will:

1. express the twelve extra rank-29 directions relative to the generic lattice;
2. enumerate low-height multisections and group them by Galois character;
3. compute same-character multiplicities and hidden product twists;
4. sieve specializations by quotient-height visibility, local solubility, and certification cost;
5. test independence incrementally by local reductions and Schur complements.

15.2 Workstream B: cubic trace-code packets

This is the main new constructive route.

1. Start from the certified split E_8 rational elliptic surface.
2. Enumerate the 2240 maximal isotropic trace codes in $E_8/3E_8$.
3. For each code, process its 240 minimal trisection classes.
4. Compute a basis of each four-dimensional linear system $|D_v|$.
5. Eliminate the surface coordinates to obtain the cubic extension of $\mathbf{Q}(t)$.

6. Require a square cubic discriminant and exactly three smooth branch values.
7. Compute the genus-one normalization and prove positive rational rank.
8. Canonicalize covers and group trisections defining the same cover.
9. Build the Hermitian Gram matrix of every group and retain rank-eleven packets.
10. Determine the ternary gluing code, projected determinant, and saturation.
11. Construct all thirty generic sections explicitly.
12. Specialize at rational points of the base and produce the final equation-and-points certificate.

A code is discarded only with a recorded exact reason: empty cyclic locus, wrong ramification, branch collision with the discriminant, base rank zero, insufficient Hermitian rank, nonintegral projected lattice, or failed descent.

15.3 Workstream C: split norm identities

Search for exact identities

$$A(x)^2 - R(x)B(x)^2 = c \prod_{i=1}^{31} (x - r_i), \quad r_i \in \mathbf{Q}.$$

Finite-field solutions are used only as lifting seeds. A candidate is promoted only when all thirty-one roots split over $\mathbf{Q}$, the corresponding points are listed exactly, and the relation module is proved to have rank one.

15.4 Workstream D: characteristic- p discovery

Reductions modulo carefully selected primes can reveal high Picard rank, character packets, and candidate trisections. They are discovery evidence, not characteristic-zero rank proofs. A configuration is promoted only after explicit lifting and exact substitution over $\mathbf{Q}(t)$.

15.5 Multi-stage sieve

For each specialization or packet, the sieve records a vector rather than one opaque score:

(forced rank, auxiliary genus, base rank, successive minima, Selmer room, coefficient height, certificate cos

Mestre–Nagao and Frobenius scores are cheap filters. They never count as mathematical evidence for rank.

15.6 Incremental independence

Given independent $P_1, \dots, P_r$ and a new point P , the pipeline:

1. verifies $P \in E(\mathbf{Q})$ exactly;
2. rejects duplicates, negatives, torsion, and known combinations;
3. tests modular dependence at good primes;
4. computes the projected quotient height;

5. updates the Gram determinant with interval-certified precision;
6. preserves the complete transformation and search provenance.

The full basis is not recomputed from scratch after every point.

15.7 Smallest decisive experiment

The next exact experiment is the minimal-shell trace-code scan:

one split E_8 surface, 2240 maximal ternary codes, 240 trisection classes per code.

The first objective is not immediately thirty sections. It is one cyclic positive-rank genus-one cover carrying at least two distinct trisection orbits. That establishes that the incidence fibres can have multiplicity greater than one. The search then escalates the Hermitian rank of the fibre toward eleven.

16 Truth-status classification and remaining uncertainty

16.1 What is proved

The repository proves the following statements unconditionally.

1. The published record curve contains twenty-nine independent rational points.
2. Thirteen independent quadratic splitting characters cannot be carried by a low-genus multiquadratic base; the minimum genus is 20481.
3. A single quadratic character can carry more than one independent Mordell–Weil direction.
4. The zeros of one rational function satisfy one forced group-law relation; a thirty-zero one-function construction spans rank at most twenty-nine.
5. Torsion-friendly rational surfaces with forced reducible fibres have insufficient packet capacity for rank thirty.
6. Elliptic fibrations with $\chi(\mathcal{O}_S) \neq 2$ are canonically rigid; the K3 case is exceptional.
7. The displayed rational elliptic surface has split Mordell–Weil lattice E_8 and twelve I_1 fibres.
8. A degree-three genus-one pullback is the first degree-three low-genus mechanism whose Hodge capacity reaches rank thirty.
9. Genus-one trisection classes are indexed by E_8 shells of norm $6(m+1)$.
10. Every cyclic trisection orbit projects to $A_2(2)$, independently of $m = D \cdot O$.
11. The minimal trace-code incidence geometry consists of 2240 maximal isotropic four-spaces, each containing 240 norm-six trisection classes.

16.2 What is not proved

No curve with thirty certified independent rational points has been produced. No cyclic trace code has yet been shown to contain eleven trisection orbits defining one positive-rank cover. Picard maximality and arithmetic descent of the proposed cubic pullback are unresolved. The degree-31 norm identity has not been solved over $\mathbf{Q}$. The project also has no universal theorem bounding rank $E(\mathbf{Q})$ by twenty-nine.

16.3 Restricted impossibility statements

The obstructions in this paper are mechanism-specific:

- arithmetic genus $d \leq 3$ over $\mathbf{P}^1$ cannot give generic rank thirty;
- four independent quadratic characters cannot be carried by a rational or elliptic multi-quadratic base;
- a thirty-zero one-function construction has a forced relation;
- ramification over an I_1 fibre prevents a cubic genus-one pullback from attaining the Hodge ceiling thirty;
- torsion-heavy rational surfaces cannot supply the required packet capacity.

None is a universal upper bound for elliptic curves over $\mathbf{Q}$.

16.4 Adversarial audit for a future Level-5 candidate

A putative rank-30 result must survive the following independent checks:

1. nonsingular global minimal model;
2. thirty complete exact rational points satisfying the equation;
3. no duplicate, negative, or torsion generators;
4. rigorous canonical-height matrix and positive determinant;
5. exact independence by finite reduction or descent;
6. saturation at every necessary prime;
7. agreement of two computer algebra systems;
8. hashes, software versions, and search provenance;
9. proof that no floating-point coordinate or cached transformation entered the certificate;
10. comparison against known curves and isomorphism canonicalization.

16.5 Current final status

The most accurate classification of this draft is

new intermediate theorem.

The cyclic cubic trace-code representation is a finite and falsifiable search space aligned with a final proof certificate. It is not yet a candidate rank- 30 construction.

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